MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. A probability distribution has probabilities summing to

   A. 0
   B. 1
   C. n
   D. the mean

2. X is 0 and 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$; $E(X) =$

   A. 1
   B. 2
   C. 3
   D. 4

3. $V(X)$ is

   A. $E(X^2)$
   B. $E(X)^2$
   C. $E(X^2) + E(X)^2$
   D. $E(X^2) - E(X)^2$

4. If $V(X) = 9$, $\sigma(X) =$

   A. 81
   B. 9
   C. 3
   D. $\frac{1}{3}$

5. $E(2X + 1)$ is

   A. $2E(X)$
   B. $2E(X) + 1$
   C. $E(X) + 1$
   D. $E(X)^2 + 1$

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain each distractor. This chapter format does not reproduce the official format of the EAM.

1. $V(X + 5)$ is

   A. $V(X) + 5$
   B. $V(X) + 25$
   C. $V(X)$
   D. $5V(X)$

2. $\sigma(-3X)$ is

   A. $3\sigma(X)$
   B. $-3\sigma(X)$

C. $9\sigma(X)$

D. $\sigma(X)$

3. The expectation is necessarily

A. a possible value

B. an integer

C. positive

D. none of these three statements

4. $E(X+Y) = E(X) + E(Y)$ requires

A. independence

B. no independence

C. X=Y

D. X positive

5. For $\sigma = 2$, n=100, $\frac{2\sigma}{\sqrt{n}} =$

A. 4

B. $0{,}04$

C. $0{,}4$

D. 2

True or false? Justify.

1. A variance may be negative.

2. Two random variables with the same expectation have the same distribution.

3. Adding a constant to a variable does not change its standard deviation.

Personal review

I build a complete distribution; I distinguish gross and net; I check both variance formulas; I do not confuse frequency with probability.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.